

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 2

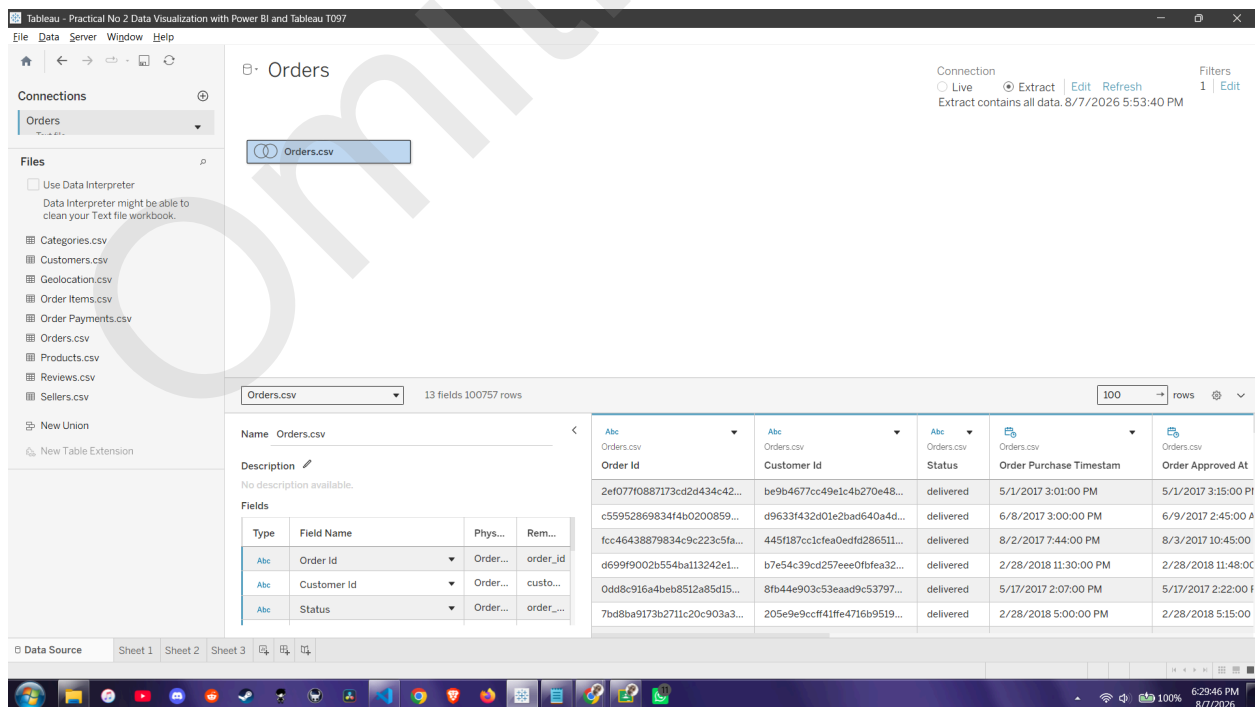
Aim: Connect to different data sources (Excel, CSV, Database)

- a) Create and use data extracts vs live connections
- b) Manage metadata (rename, hide, change data types)
- c) Perform joins (inner, left, right) on multiple tables
- d) Apply data blending between two sources
- e) Implement different types of filters (dimension, measure, date)
- f) Optimize performance using extracts and filtering

A. Connect to different data sources

- Launch the Tableau Desktop application.
- Under the blue Connect panel on the left side of the start page, click on Text file.
- Navigate to your local directory, select Orders . csv, and click Open.

After performing the following steps this is what you will be seeing:

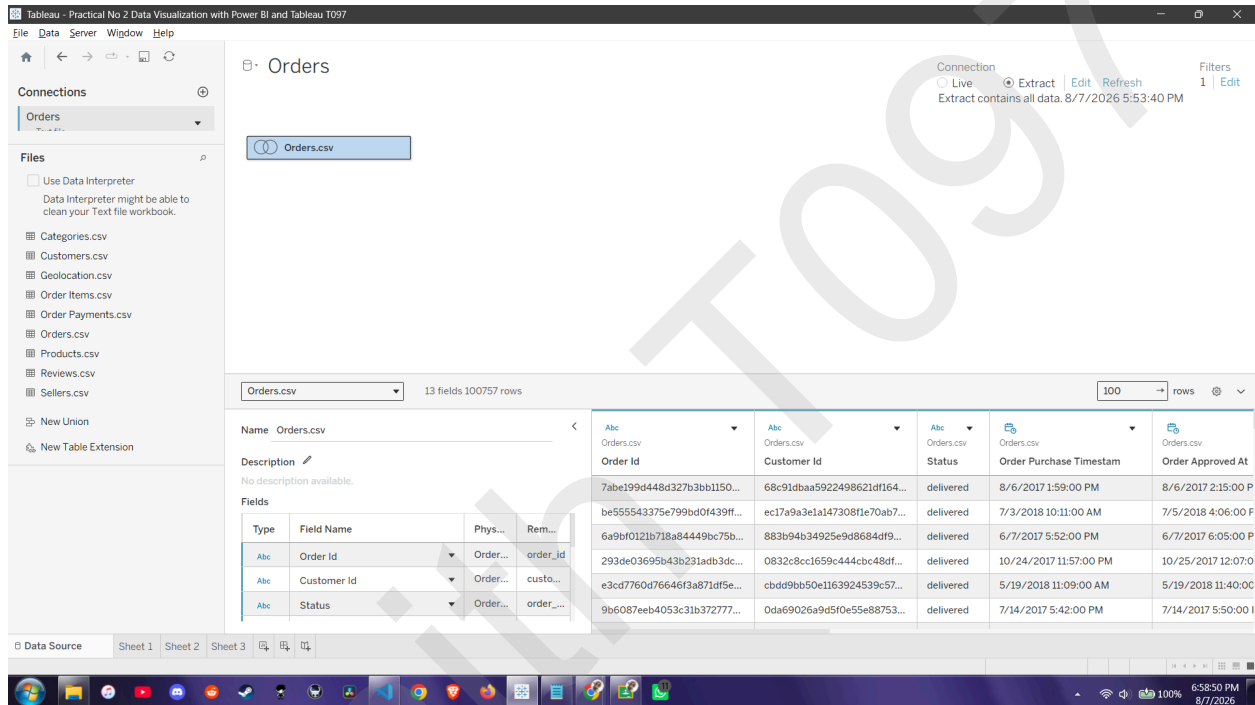


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B. Create and use data extracts vs live connections

- In the top-right corner of the Data Source interface, locate the connection mode toggle.
- Switch the active selection from Live over to Extract.
- Click on the Sheet 1 tab at the bottom-left corner of the workspace.
- A 'Save Extract As' dialog box will appear. Name the local snapshot file Ecommerce_Extract and click Save.

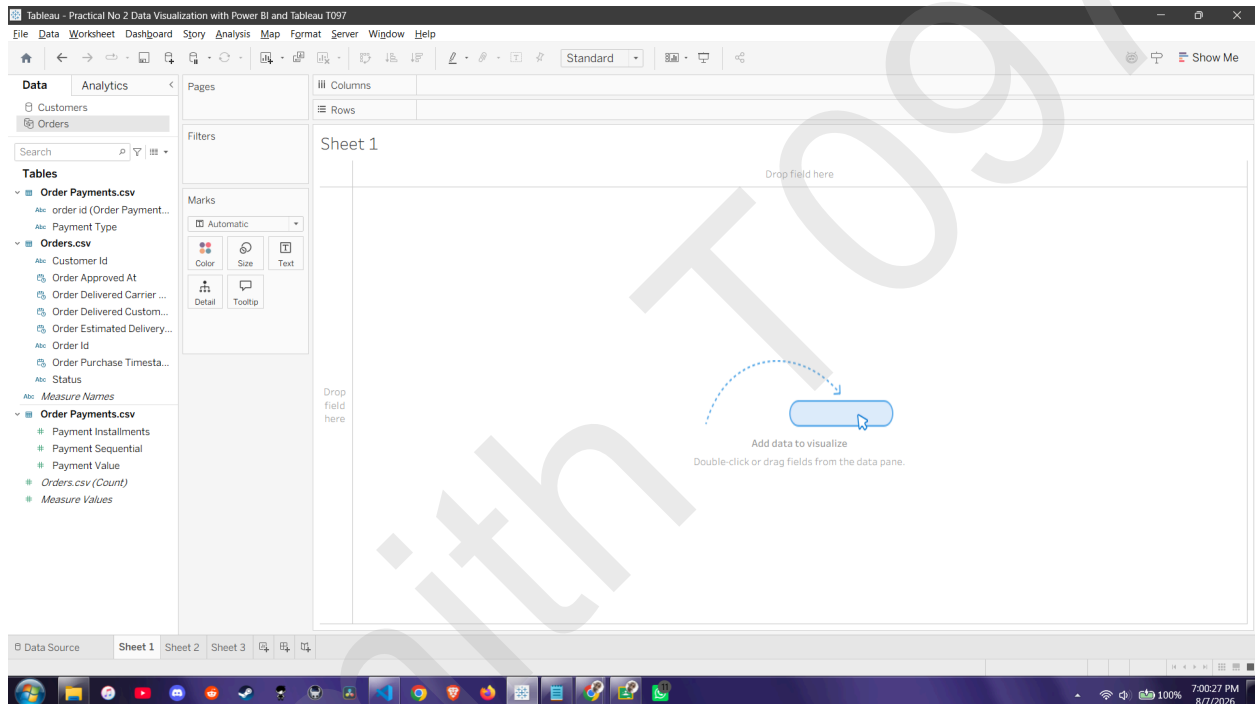
After performing the following steps this is what you will be seeing:



C. Manage metadata (rename, hide, change data types)

- On Sheet 1, locate the field currently labeled `order_status` in the Data Pane. Right-click the field name, select **Rename**, type in `Status`, and hit Enter.
- Locate the `customer_id` dimension. Right-click the field name and select **Hide**.
- Locate the `order_purchase_timestamp` column. Click directly on the data type icon (it may display as text `Abc`) and select **Date & Time** from the dropdown menu to correctly format it.

After performing the following steps this is what you will be seeing:



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D. Perform joins (inner, left, right) on multiple tables

- Click the **Data Source** tab at the bottom-left to return to the schema designer.
- Double-click directly on the rectangular **Orders . csv** block to drill down into the Physical Layer canvas space.
- From the left-hand files listing pane, click and drag the **Order Payments . csv** file, dropping it directly next to the active **Orders . csv** block.
- Click directly on the Venn diagram icon to open the Join configuration box. Set **order _id** as the matching key for both tables.
- Leave the configuration set to a **Left Join** to safely preserve all root rows from **Orders . csv**.

After performing the following steps this is what you will be seeing:

The screenshot shows the Tableau Desktop interface with the 'Orders' data source selected. The 'Data Source' tab is active at the bottom. The 'Orders.csv' table is selected, and the 'Order Payments.csv' table is being added to it. The 'Join' configuration box is open, showing a Venn diagram and the join type set to 'Left Join'. The 'Fields' pane on the left lists the fields for 'Orders.csv' and 'Order Payments.csv'.

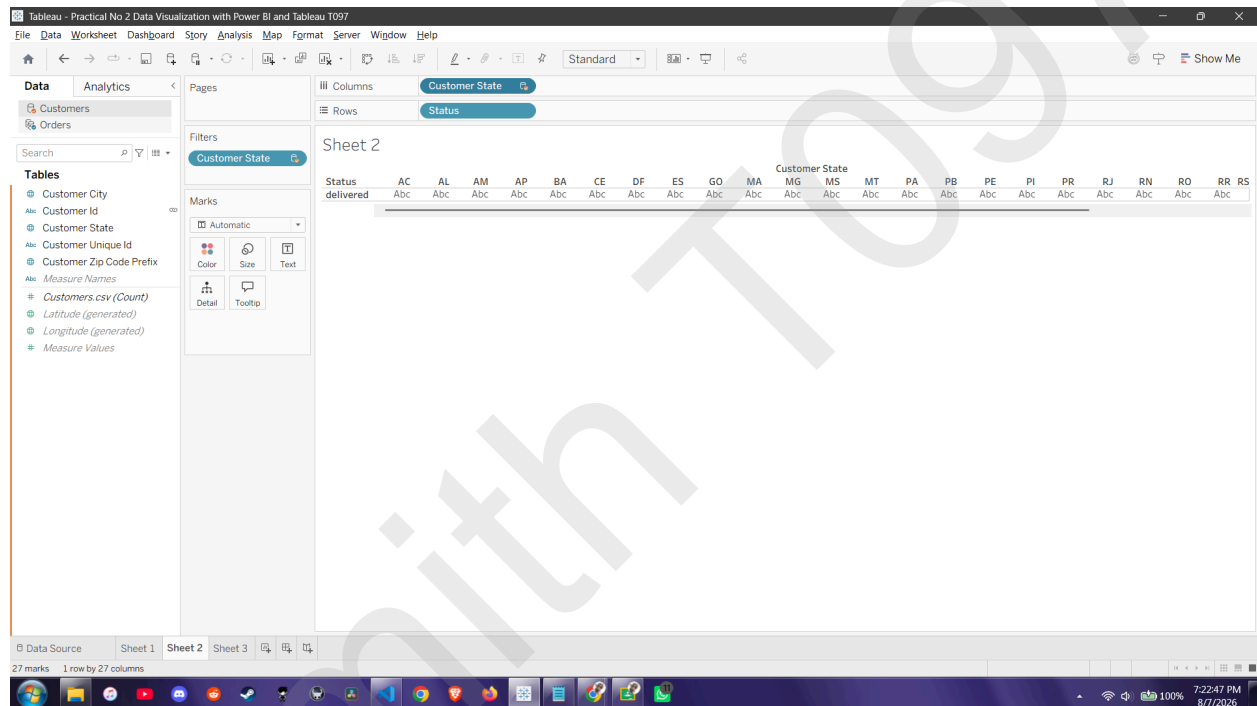
Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approved At
7abe199d448d327b3bb1150...	68c91dbaa5922498621df164...	delivered	8/6/2017 1:59:00 PM	8/6/2017 2:15:00 P
be555543375e799bd0f439ff...	ec17a9a3e1a1473081e70ab7...	delivered	7/3/2018 10:11:00 AM	7/5/2018 4:06:00 F
6a9bf0121b718a84449bc75b...	883b94b34925e9d8684df9...	delivered	6/7/2017 5:52:00 PM	6/7/2017 6:05:00 P
293de03695b43b231adb3dc...	0832c8cc1659c444cbc48df...	delivered	10/24/2017 11:57:00 PM	10/25/2017 12:07:0
e3cd7760d76646f3a871df5e...	cbdd9bb50e1163924539c57...	delivered	5/19/2018 11:09:00 AM	5/19/2018 11:40:0C
9b6087eeb4053c31b372777...	0da69026a9d5f0e55e88753...	delivered	7/14/2017 5:42:00 PM	7/14/2017 5:50:00 I

E. Apply data blending between two sources

- Create a new worksheet (Sheet 2). Drag the Status dimension to the **Rows** shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.
- Go to **Data** $\rightarrow$ **New Data Source** $\rightarrow$ **Text file** and add Customers.csv.
- Navigate to **Data** $\rightarrow$ **Edit Blend Relationships**.
- Set Primary to Orders and Secondary to Customers. Switch from Automatic to **Custom**, click **Add**, pair customer_id from the left list to customer_id in the right list, and click OK.
- In the Data Pane, click the Customers.csv dataset and click the broken link icon next to customer_id so it turns into a solid orange chain link.

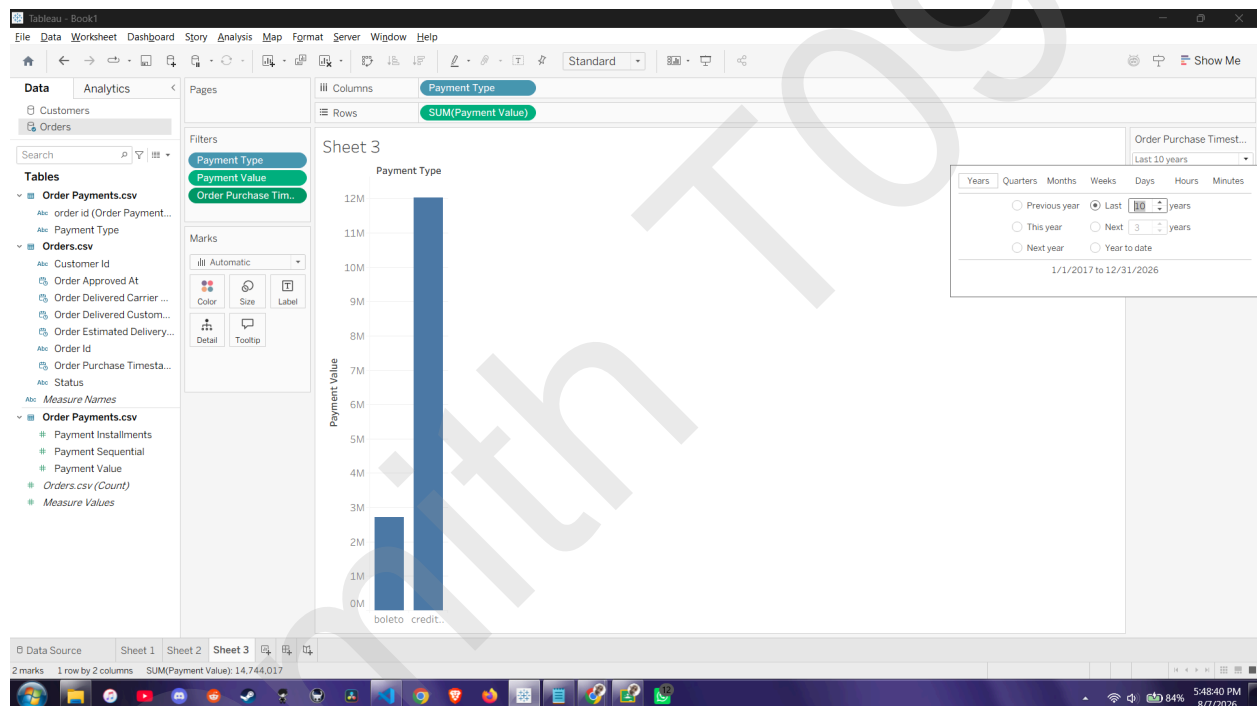
After performing the following steps this is what you will be seeing:



F. Implement different types of filters (dimension, measure, date)

- Open a new worksheet (Sheet 3). Drag payment_value to the **Rows** shelf and payment_type to the **Columns** shelf.
- **Dimension Filter:** Drag payment_type to the **Filters** shelf. Uncheck not_defined, check credit_card and boleto, and click OK.
- **Measure Filter:** Drag payment_value to the **Filters** shelf. Select **All Values**, set the condition to **At Least**, type in 50, and click OK.
- **Date Filter:** Drag order_purchase_timestamp to the **Filters** shelf. Select **Relative Date** or **Years** to isolate specific historical boundaries, and click OK.

After performing the following steps this is what you will be seeing:



G. Optimize performance using extracts and filtering

- Return to the **Data Source** tab. Below the Extract radio dial in the upper-right, click **Edit**.
- Click **Add**, select the Status field, check delivered exclusively, and click OK to purge unfulfilled order rows from the local extract.
- Return to an active worksheet. Right-click the date filter pill on your Filters shelf and select **Add to Context**. The pill will turn dark gray, forcing Tableau to process it first.

After performing the following steps this is what you will be seeing:

